ROBO STEAM ACADEMY — Teachers' Toolkit

Where to download the PicoBot Teachers' Toolkit in five languages

1. The toolkit

Teachers' ST(R)EM: Robots in STEM Toolkit — everything a teacher needs to teach robotics and programming with **PicoBot**, a small mecanum-wheel robot with an arm, built on the **Raspberry Pi Pico 2 W / Pico W** and programmed in **MicroPython**: lesson plans for 32 double sessions (full course) or 10 sessions (short course), student materials, slide decks, quizzes, reference documents, methodological guides and all lesson programs.

Every document is available as **PDF** (to read and print) and **Word** (to adapt).

2. Download

Each language is a shared folder on the project's cloud. Open the link and use the **Download** button to get the whole folder, or open single documents in the browser.

Language	Link
English	https://cloud.robosteam.eu/s/KQbdqP544KEqqYk
Български (Bulgarian)	https://cloud.robosteam.eu/s/c93YNFNpZ6drfjL
Română (Romanian)	https://cloud.robosteam.eu/s/cCCc8LCmyj7Ac2Z
Slovenčina (Slovak)	https://cloud.robosteam.eu/s/YFi9EA8Td4mY4JH
Українська (Ukrainian)	https://cloud.robosteam.eu/s/6PgoN8ZanzJo4dD

> **Note**
>
> The folders always contain the current version of the toolkit; the version number is printed in the footer of every document. English is the master version. The Bulgarian, Romanian, Slovak and Ukrainian editions were translated with the help of artificial intelligence; native speakers of the partner countries check them.

3. Start here

1. Read the **Teacher's Guide** (folder `teachers/0_start_here/`): course plans, classroom organisation, safety.
2. Build and set up the robots with **A2** Assembly and maintenance and **A3** Software set-up (folder `teachers/3_reference/`).
3. Teach with the **teacher edition** of each module (`teachers/1_modules/`) and its slide deck (`teachers/2_slides/`). The student materials are in `students/`, the lesson programs in `code/`.

4. Videos and robot programs

Seven videos with subtitles in English, Bulgarian, Romanian, Slovak and Ukrainian (press **CC** on YouTube):

Video	Link
V1 PicoBot assembly - build the robot step by step	https://www.youtube.com/watch?v=kURZ2_49MeE
V2 PicoBot demo - mecanum moves and arm control	https://www.youtube.com/watch?v=Bz0MArCKrIQ
V3 PicoBot object manipulation	https://www.youtube.com/watch?v=MrGmpOJrRMA
V4 PicoBot garbage collection	https://www.youtube.com/shorts/GdaHDtyYFLE
V5 PicoBot obstacle avoidance	https://www.youtube.com/shorts/-lE8YsaX5aQ
V6 PicoBot line following	https://www.youtube.com/shorts/pWShT70neII
V7 Competitions Platform - Participant Guide	https://www.youtube.com/watch?v=o5tVkmkcWV4

The PicoBot library and the ready robot programs: https://github.com/robosteamdev

This page online: https://github.com/robosteamdev/robo-steam-academy-teachers-toolkit

About this toolkit



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Partners: National STEM Center (coordinator, Bulgaria, https://stem.mon.bg/) · Institute of Information and Communication Technologies, Bulgarian Academy of Sciences (Bulgaria, https://www.iict.bas.bg/) · Academy Nikola Tesla (Bulgaria, https://ant.bg/) · Technical University of Košice (Slovakia, https://www.tuke.sk/) · Stredná priemyselná škola elektrotechnická Prešov (Slovakia, https://spseadlerka.edupage.org/) · National Association of Public Librarians and Libraries in Romania – ANBPR (Romania, https://anbpr.org.ro/) · Liceul Tehnologic Cezar Nicolau (Romania, https://www.cezar-nicolau.ro/).

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